Written By: Tony Cruise

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In this section I will explain how you attach extra commands to ROM routines.

In MSX computers there are many Hook Jumps provided so that machine code programmers can attach their own routines to most of the basic ROM routines. Each Hook consists of five memory locations in RAM. When the computer is first switched on or reset, each of the locations contains the value 201.

This is the decimal value for a RETURN statement in machine code.

Each of the Hooks are jumped to at the start of each of the corresponding ROM routines, so you can insert or eliminate entirely each routine.

To use a Hook, all you have to do is insert your own commands into the five memory locations.

This is usually a JUMP instruction to your machine code routine.

Here is an example program that uses the Hook HTIMI. This Hook is jumped to every 1/50th of a second.

Machine Code Program

D000		1 ORIGIN	ORG 0D000H	
		2 ;		
		3 ; PROGRAM TO UPDATE AND DISPLAY A CLOCK		
		4 ; ON SCREEN 0		
		5 ;		
		6 ; LABEL SETTINGS		
		7 HTIMI	EQU 0FD9FH	; Hook Jump – Timer
		8 SCRMOD	EQU 0FCAFH	; RAM Storage – Screen Mode
		9 SETWRT	EQU 0053H	; ROM Routine – Set Video Write
		10 ;		
D000	3EC3	11 START:	LD A,0C3H	; Value for JUMP instruction
D002	329FFD	12	LD (HTIMI),A	; Load into Hook
D005	210CD0	13	LD HL,CUE	; Address to jump to

D008	22A0FD	14	LD (HTIMI+1),A	; Load into Hook
D00B	C9	15	RET	
		16 ;		
D00C	CD14D0	17 CUE:	CALL CLOCK	; Call CLOCK routine
D00F	F7	18	RST 030H	; Disk delay routine
D010	879C77	19	DEFB 087H,09CH,077H	; remove if you do not
D013	C9	20		; have a disk drive
		21 ;		
D014	161D0	22 CLOCK:	LD HL,STORE	; Decrease counter
D017	7E	23	LD A,(HL)	;
D018	3D	24	DEC A	;
D019	77	25	LD (HL),A	;
D01A	C0	26	RET NZ	; Return if not ZERO
D01B	3E33	27	LD A,51	; Set new counter
D01D	77	28	LD (HL),A	;
D01E	2162D0	29	LD HL,CLKSTR	; Update – seconds
D021	0603	30	LD B,3	;
D023	7E	31 CL1:	LD A,(HL)	;
D024	3C	32	INC A	;
D025	27	33	DAA	; Decimal addition
D026	FE69	34	CP 060H	;
D028	3003	35	JR NC,CL2	;
D02A	77	36	LD (HL),A	;
D02B	1805	37	JR PRTCLK	;
D02D	3600	38 CL2:	LD (HL),0	;
D02F	23	39	INC HL	;
D030	10F1	40	DJNZ CL1	;
D032	3AAFFC	41 PRTCLK:	LD A,(SCRMOD)	; Get screen mode
D035	3D	42	DEC A	;
D036	D0	43	RET NC	; Check for screen 0
D037	211F00	44	LD HL,31	;
D03A	CD5300	45	CALL SETWRT	; Set screen address
D03D	2164D0	46	LD HL,CLKSTR+2	; Set pointer to clock
D040	0603	47	LD B,3	; Set counter
D042	7E	48 P1:	LD A,(HL)	;
D043	27	49	DAA	;
D044	F5	50	PUSH AF	; Save value
D045	CB3F	51	SRL A	; Get right digit
D047	CB3F	52	SRL A	;
D049	CB3F	53	SRL A	;
D04B	CB3F	54	SRL A	;
D04D	C630	55	ADD A,48	; Change to ASCII
D04F	D398	56	OUT (098H),A	; Send to screen
D051	F1	57	POP AF	; Return value

D052	E60F	58	AND 1111B	; Get left digit
D054	C630	59	ADD A,48	; Change to ASCII
D056	D398	60	OUT (098H),A	; Send to screen
D058	3E3A	61	LD A,58	; Value for ‘:’
D05A	2B	62	DEC HL	; Update pointer
D05B	05	63	DEC B	; Update counter
D05C	C8	64	RET Z	; Return if finished
D05D	D398	65	OUT (098H),A	; Send ‘:’ to screen
D05F	18E1	66	JR P1	; Go again
		67 ;		
		68 ; DATA STORAGE		
D061	33	69 STORE:DEFB 51		
D062	000000	70 CLKSTR:DEFB 0,0,0		
D065		71 END:		

Basic Loader

```
10 CLS:CLEAR 200,&HCFFF:DEFINTA-Z:A=&HD000
20 READ A$:IF A$ <> “@” THEN POKE A, VAL(“&H”+A$):A=A+1:GOTO 20
30 PRINT”  INSERT TAPE/DISK TO SAVE PROGRAM”
40 PRINT”       AND PRESS ANY KEY”
50 A$=INPUT$(1):PRINT:PRINT “  SAVING ....”
60 BSAVE”CLOCK”,&HD000,A-1
100 DATA 3E,C3,32,9F,FD,21,0C,D0,21,0C,D0,22,A0,FD,C9,CD
110 DATA 14,D0,F7,87,9C,77,C9,21,61,D0,7E,3D,77,C0,3E,33
120 DATA 77,21,62,D0,06,03,7E,3C,27,FE,60,30,03,77,18,05
130 DATA 36,00,23,10,F1,3A,AF,FC,3D,D0,21,1F,00,CD,53,00
140 DATA 21,64,D0,06,03,7E,27,F5,CB,3F,CB,3F,CB,3F,CB,3F
150 DATA C6,30,D3,98,F1,E6,0F,C6,30,D3,98,3E,3A,2B,05,C8
160 DATA D3,98,18,E1,33,00,00,00,@
```

If you do not have a disk drive, change line 110 to -;

110 DATA 14,D0,**00,00,00,00**,C9,21,61,D0,7E,3D,77,C0,3E,33

This program displays a clock in the top right hand corner of the screen, but only when the computer is in screen mode zero. The first part of the program fills the Hook HTIMI with a jump command pointing to a routine called CUE. This routine calls all the things that are using this Hook.

In this case, just the two things are being called, the clock routine, and the routine to turn the disk drive motor off after a certain period of time.

If you are not using a disk drive, set the four memory locations to zero.

In later articles, I will detail more routines that use this Hook.

The clock routine starts by decreasing the first counter, which was initially set to 51. This counter is to measure the passing of one second (The Hook is called every 1/50th of a second).

Once a second has passed, the clock counters for the hour, minutes and seconds are updated.

Then, if the screen mode is set to zero, the clock will be printed out.

In the next section, I will cover another of the Hook jumps called **HGONE**, and show how new statements can be added to your Basic programs.